

VERIFIED RESULTS · REFRIGERATED DISTRIBUTION CENTRE

Patented capital equipment cut refrigeration energy

26.5% at a refrigerated distribution centre.

Validated by ON/OFF metering. Sub-12-month payback. Zero cold-chain disruption.

26.5%	112,432	£27,936/yr	~11mo
REFRIGERATION REDUCTION	KWH ANNUALIZED	FINANCIAL BENEFIT	PROJECTED PAYBACK

SOLUTION Patented enPact Refrigeration & Cooling optimization. Installed across all 5 cold rooms and 3 freezer systems. Reduces compressor runtime (80–90% of refrigeration energy) via superheat-based control and stabilizes setpoints without modifying site operations. - Reduce compressor runtime and cycling - Regulate refrigerant flow via superheat control - Increase latent cooling efficiency - Maintain setpoint temperatures without modification - Verified via smart kWh meters + M&V (Univers platform)	CHALLENGE A national food distribution center with a large refrigerated warehouse faced rising energy consumption. Refrigeration represented the majority of the site’s electrical usage. Key constraints: - Freezers operating at 68.5 kW each; cold rooms 15–19 kW 25% seasonal uplift during warmer months - Transparent, M&V-validated savings required before wider rollout - Sub-12-month payback required for investment approval - No option for cold-chain disruption
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Results. Whole-warehouse refrigeration savings. Measured over 34 days (Feb–Mar) using ON/OFF comparison windows. Full-year figure adjusts for the 25% summer uplift documented onsite. 26.5% total reduction in refrigeration energy usage 219+ kWh/day saved across freezer rooms alone 58 kWh/day saved across cold rooms 101,126 kWh saved on measured pilot basis 112,432 kWh full-year (seasonally adjusted) - Sub-12-month payback achieved	VERIFIED METRICS <table><tr><td>Technology</td><td>enPact R&C</td></tr><tr><td>Refrigeration reduction</td><td>26.5%</td></tr><tr><td>Annualized savings</td><td>112,432 kWh</td></tr><tr><td>Financial benefit</td><td>£27,936/yr</td></tr><tr><td>CO₂ reduction</td><td>66,335 kg/yr</td></tr><tr><td>Payback</td><td>~11 months</td></tr></table>	Technology	enPact R&C	Refrigeration reduction	26.5%	Annualized savings	112,432 kWh	Financial benefit	£27,936/yr	CO ₂ reduction	66,335 kg/yr	Payback	~11 months
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Measurement & verification. All equipment monitored via smart kWh meters feeding the Univers platform. Winter measurement was seasonally adjusted upward using the site’s documented 25% summer refrigeration uplift to produce the full-year 112,432 kWh figure. No refrigerant, setpoint, or workflow changes were made during the trial.

CONCLUSION enPact Refrigeration & Cooling delivered a validated 26.5% reduction across a large refrigerated warehouse, ~11-month payback, and material CO₂ abatement — with zero cold-chain disruption.	ESTATE ROLLOUT PROJECTION Applied across 340 similar sites: 34.4M kWh/yr savings, £8.25M/yr financial benefit, 7,119 tons CO₂/yr abated, ~12-month portfolio payback.
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